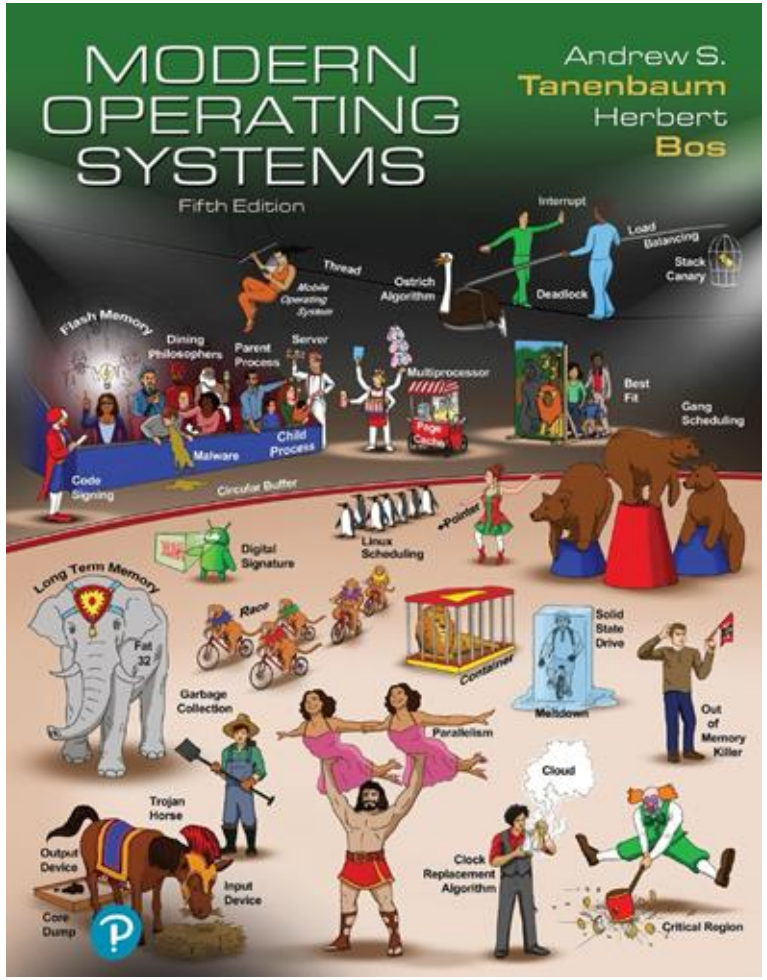
Modern Operating Systems

Fifth Edition



Memory Management

Pearson

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Memory

- Memory Abstractions
- Virtual Memory
- Page Replacement Algorithms
- Design Issues for Paging Systems

Virtual Memory (1)

- There is a need to run programs that are too large to fit in memory
- Solution adopted in the 1960s, split programs into little pieces, called overlays
  - Kept on the disk, swapped in and out of memory
  - Initially, only the overlay manager (O M) was loaded
  - The O M would then load overlay 0 and run it
  - When done, it would tell the O M to load overlay 1, etc.
- Nowadays, we have a better solution: *virtual memory*
  - Often implemented using *paging*

Virtual Memory (2)

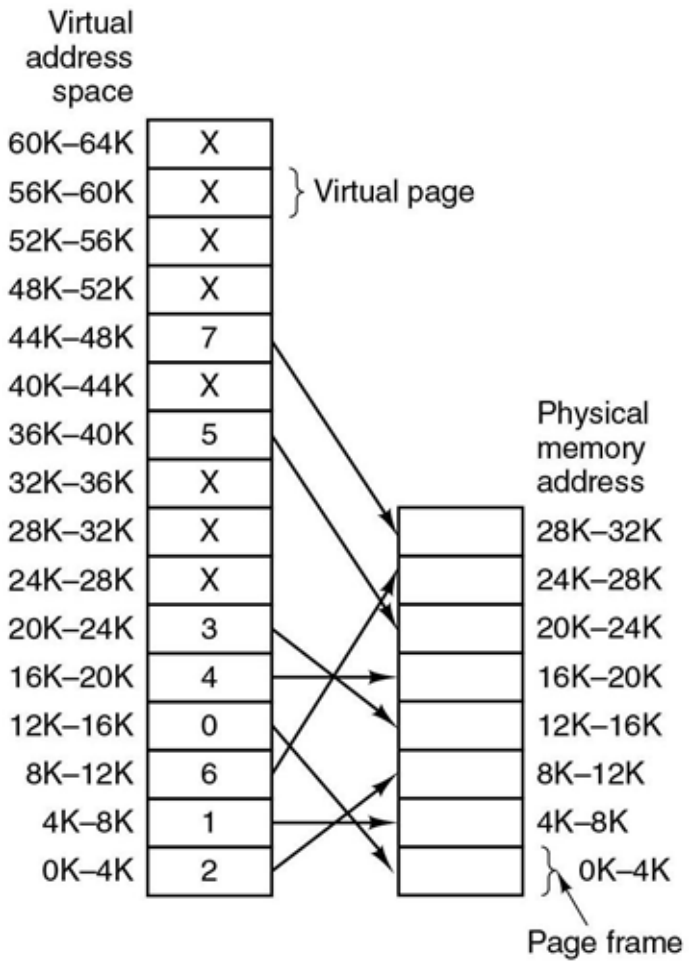
Problem: So far, memory allocation to processes has been restricted to contiguous chunks. This means that a process must be loaded into a single, uninterrupted block of available memory, which can lead to inefficiencies like external fragmentation (wasted space between allocated blocks).

Solution:

- Establish the illusion of a vast address space for the process.
  - e.g., 32-bit or 48-bit (16-bit address space in our example)
- This construct is termed the **virtual address space**.

Practical Issue:

- The actual physical memory is significantly constrained
  - Translate the virtual addresses employed by the process into corresponding physical addresses.
  - This translation is routinely executed by the Memory Management Unit (MMU).



Modern systems use paging: Divide physical and virtual memory into pages of fixed size (e.g. 4096 bytes)

Virtual to Physical Memory Address Translation

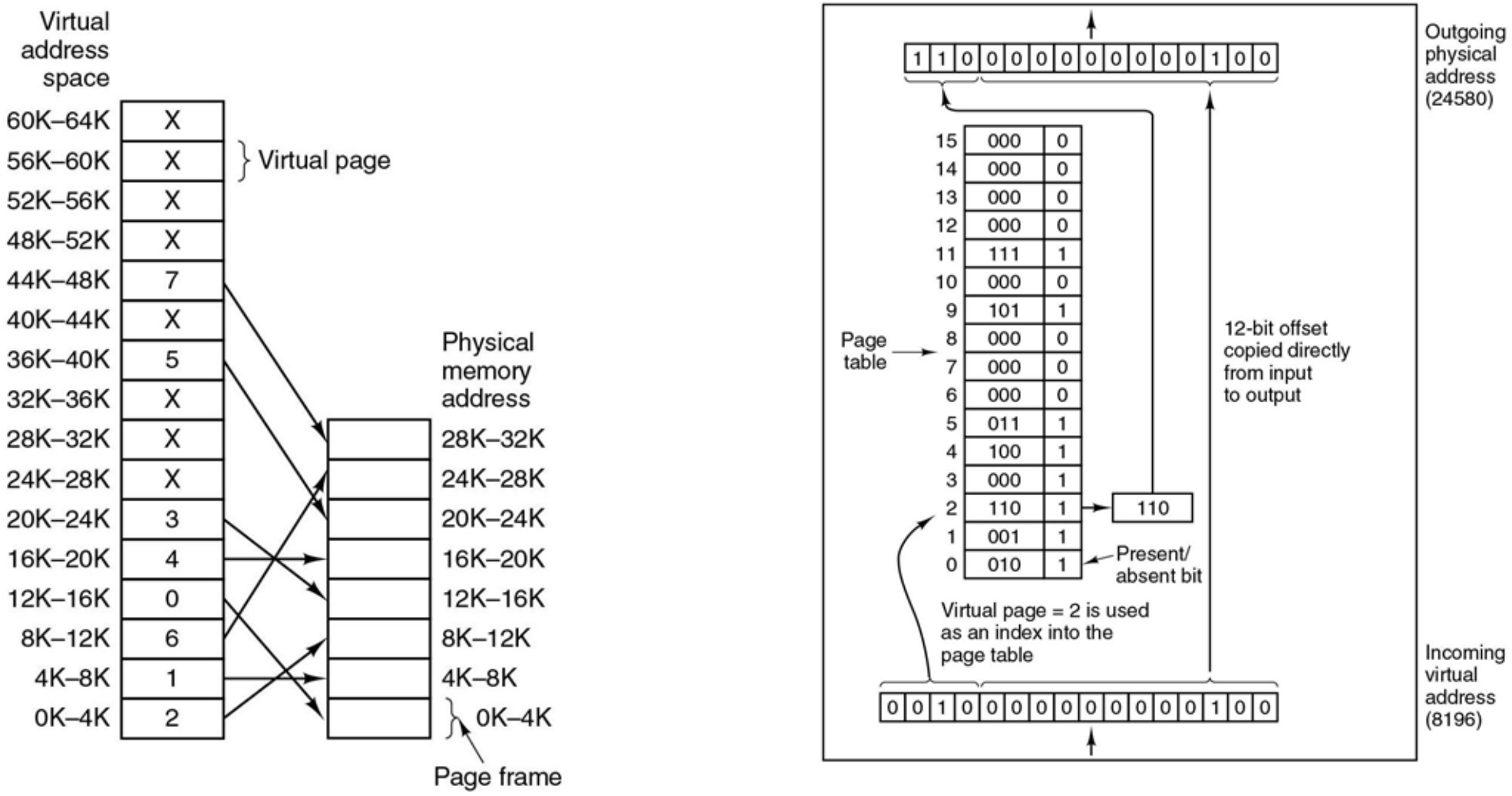
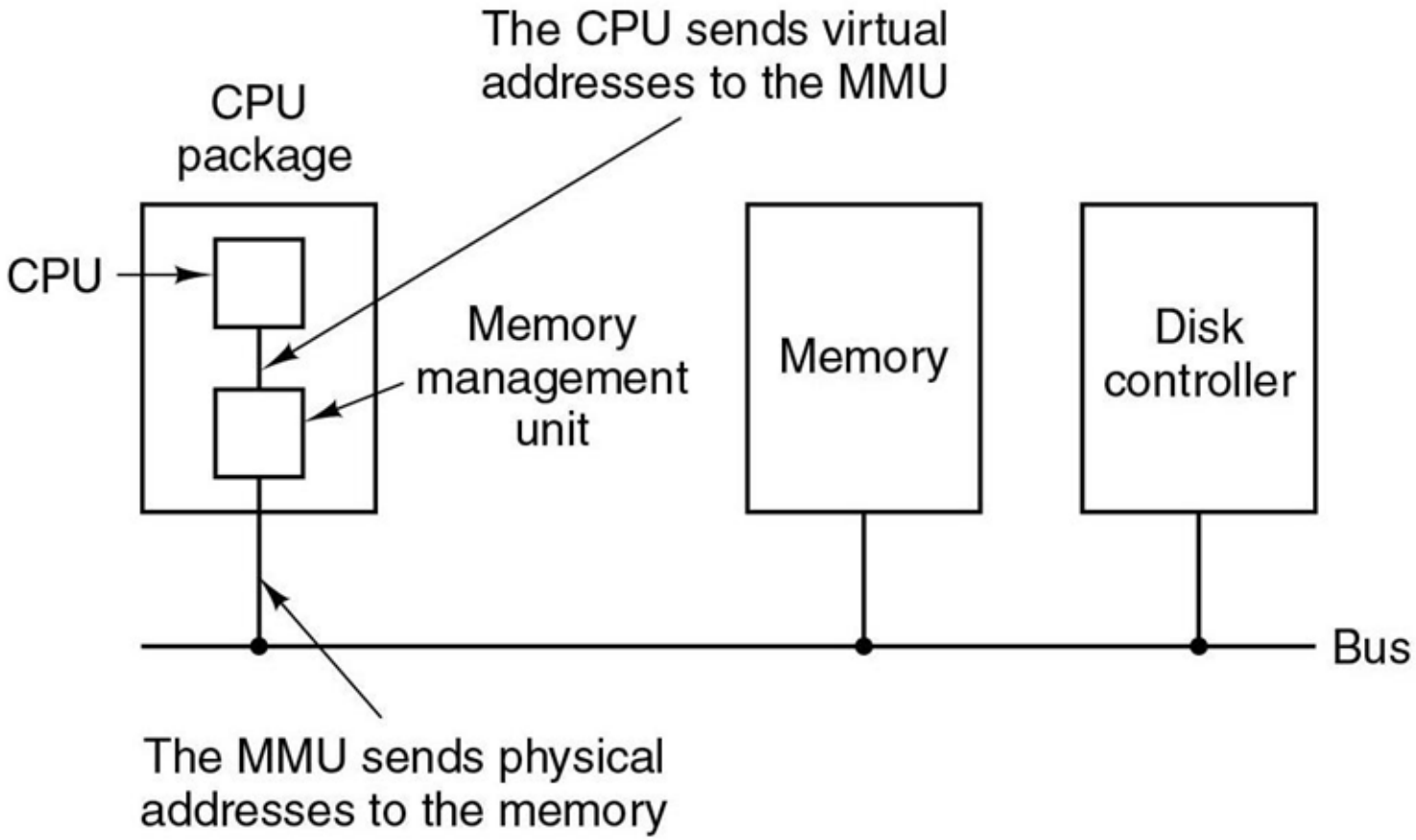


Figure 3.10 The internal operation of the MMU with 16 4-K B pages.

Memory Management Unit



Question: where we store page table of process?

Structure of a Page Table Entry

Questions:

- How big is the page table?
- How many page tables?

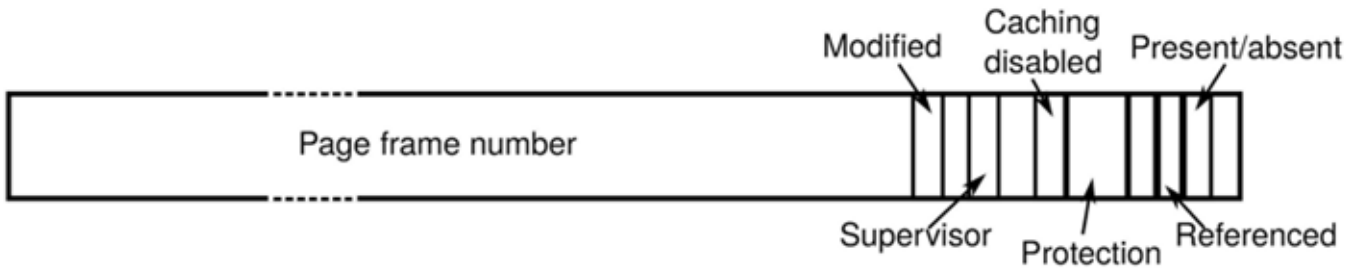


Figure 3.11 A typical page table entry.

- 4KB page size (typical)
- 32 bit systems:
  - 32 bit address space and up to 2^(32-12) page table entries (PTEs)
  - 4-byte per PTE, thereby up to 4 MB each process.
- 64 bit systems
  - Really: 48-bit or 57-bit address space.
  - Question: How many PTEs in if address space is 48 bit?

Page Table Sizes

A very large virtual address space would lead to a very large page table $\rightarrow$ not nice!

Possible solutions:

- Multi-level page tables
- Inverted page tables

Two-level Page Tables (x86)

Page tables are “walked” by hardware
MMU = Memory Management Unit

CR3 register
(special register to point to top of page table hierarchy)

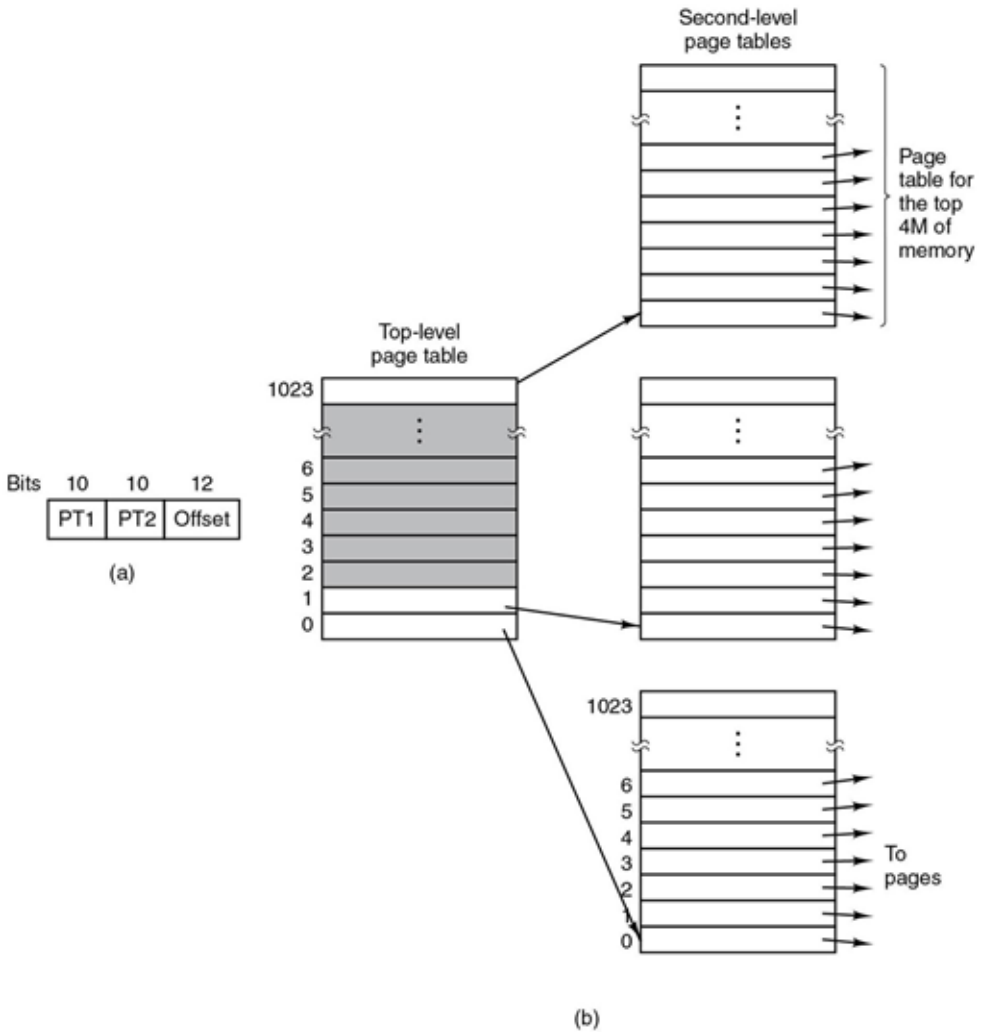
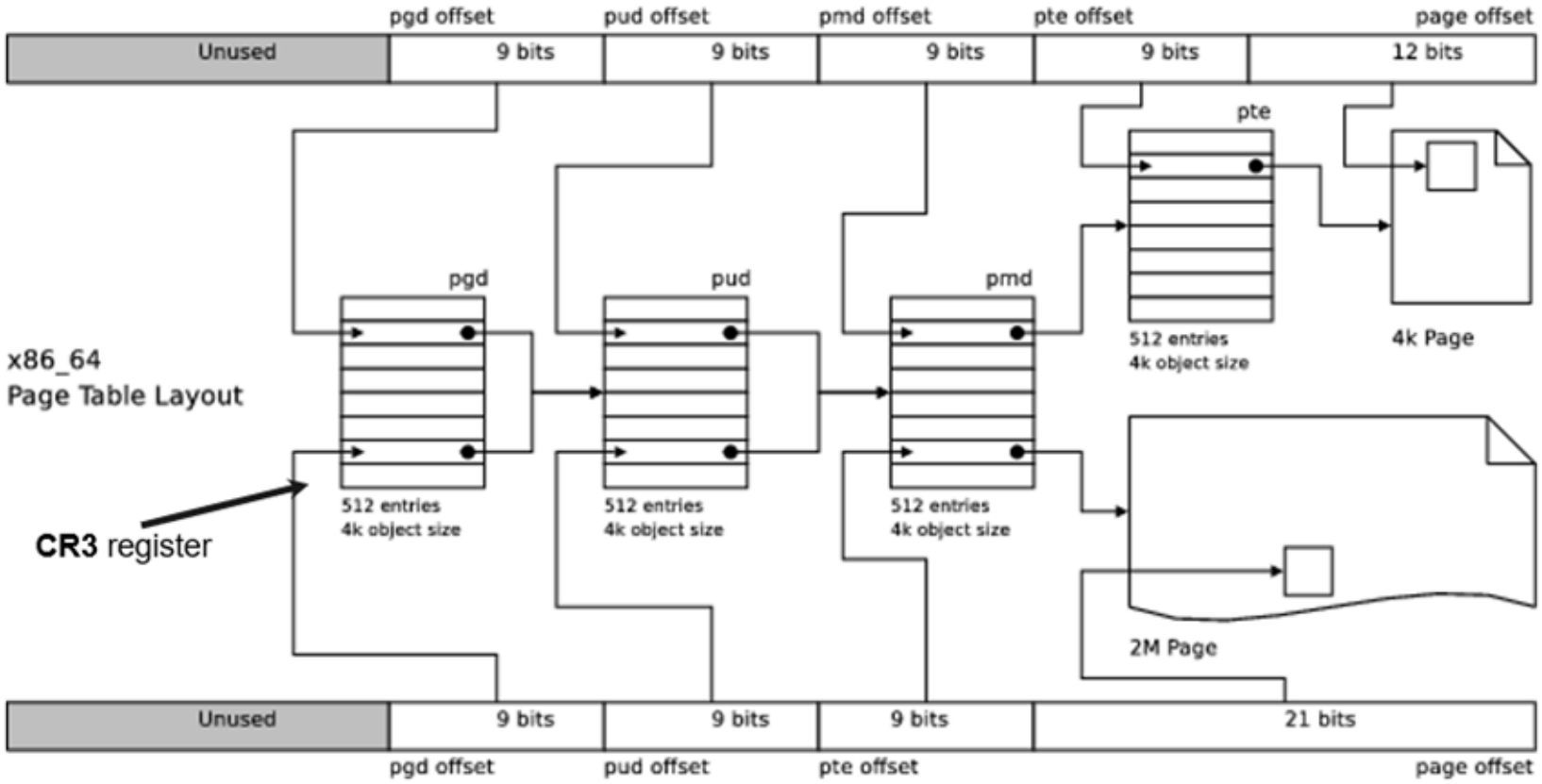


Figure 3.13 (a) A 32-bit address with two page table fields. (b) Two-level page tables.

– Question: How big is the table if three second-level page tables are active.

Four-level Page Tables (x86-64)



PGD: Page Global Directory, **PUD:** Page Upper Directory **PMD:** Page Mid-level Directory, **PTE:** page table entry

Inverted Page Tables (I A-64)

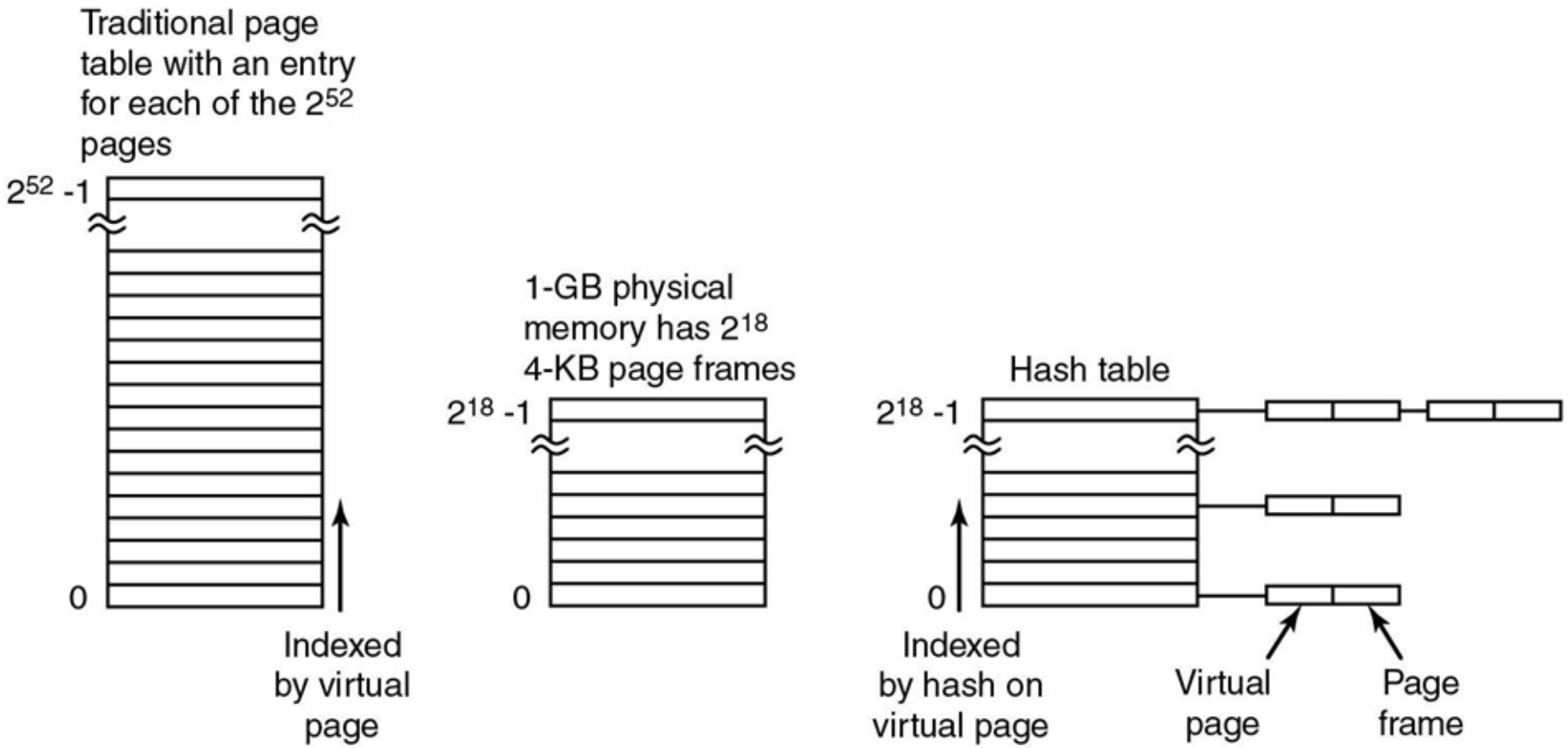
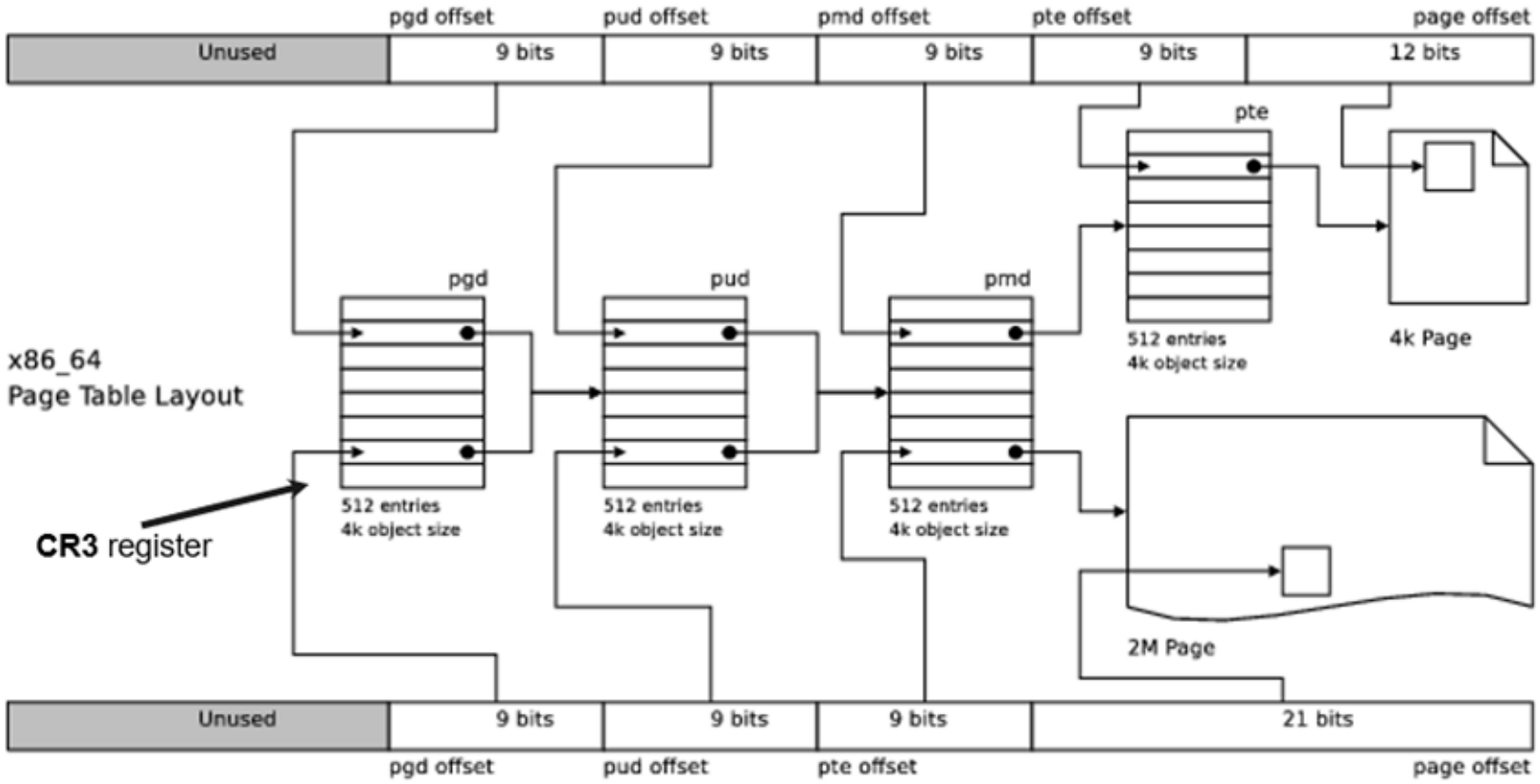


Figure 3.14 Comparison of a traditional page table with an inverted page table.

Four-Level Page Tables: Lots of Memory Accesses!



PGD: Page Global Directory, **PUD:** Page Upper Directory, **PMD:** Page Mid-level Directory, **PTE:** page table entry

Translation Lookaside Buffer (1)

Problem: MMU has to translate every single memory access

→ Performance problem

Solution: Cache previous translations, hoping programs

→ A high degree of locality.

Valid	Virtual Page	Modified	Protection	Page Frame
1	140	1	RW	31
1	20	0	R X	38
1	130	1	RW	29
1	129	1	RW	62
1	19	0	R X	50
1	21	0	R X	45
1	860	1	RW	14
1	861	1	RW	75

Figure 3.12 A T L B to speed up page.

Translation Lookaside Buffer (2)

Extra material

Questions:

- How many T L B s?
- When to flush T L B?
- When to expect many T L B misses?

Translation Lookaside Buffer (3)

Extra material

Questions:

- **How many TLB s?**
- When to flush TLB?
- When to expect many TLB misses?

1. Super expensive memory: small number of entries (e.g., 64-128)
2. Sometimes separate entries for huge (2 MB) pages and normal (4 KB) pages
3. Often separate TLB s for code and data
4. 1 per CPU, but sometimes multiple levels of TLB (L1 TLB vs L2 TLB)

Translation Lookaside Buffer (4)

Extra material

Questions:

- How many TLB s?
- **When to flush TLB?**
- When to expect many TLB misses?

1. When entries change
2. Typically at context switch

Translation Lookaside Buffer (5)

Extra material

Questions:

- How many TLBs?
- When to flush TLB?
- **When to expect many TLB misses?**

1. After flushes
2. When the program exhibits little locality

TLB Misses and Page Fault Handling (1)

Software- vs. hardware-managed TLB:

- TLB misses handled by OS vs. hardware → Hardware walks PTs and fills TLB

  ↓

  E.g., OS may preload TLB entries that it expects to need later

- Flexibility vs. Efficiency

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